RV College of Engineering®

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Academic year 2025-2026 (Odd Sem)

DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING

Date	November 2025	**Maximum Marks**	50 + 10
Course Code	CS3721A	**Duration**	90+20Minutes
Sem	7^{th}		

PARALLEL ARCHITECTURE AND GPU PROGRAMMING

CIE-I

Sl. No.		**PART-A**	M	BT	CO
1.1		Identify and justify a primary quantitative principle in computer design that is used to improve performance.	02	L3	1
1.2		Mention a usecase/instruction sequence in which the locality of reference concept would fail.	02	L3	1
1.3		Justify the purpose and need of speculative execution in computer architecture.	02	L2	3
1.4		If an exception is raised and the succeeding instructions are executed completely, then the processor is said to have ______________.	01	L3	1
1.5		Which compiler technique increases ILP by reducing the overhead of loop control and allowing instructions from different iterations to be scheduled together?	01	L2	1
1.6		The Tomasulo algorithm uses what hardware component to hold instructions waiting for their operands to become available, allowing out-of-order execution?	01	L1	3
1.7		What is the key bottleneck in a Symmetric Shared Memory (UMA) architecture?	01	L2	3
		PART-B			
1		Define computer architecture with ISA and explain seven dimensions of ISA.	10	L2	1
2	(a)	Illustrate and explain with an example datadependence in pipeline that requires stalls. Write its MIPS code snippet.	06	L3	3
	(b)	Considering the pipelined execution of the following instructions, identify the various types of dependencies existent between these instructions. DADD R11,R21,R31 DSUB R41,R11,R51 AND R61,R11,R71 OR R81,R11,R91 XOR R10,R11,R12	04	L3	3
3	(a)	Assume a disk subsystem with the following components and MTTF: ■ 10 disks, each rated at 1,000,000-hour MTTF ■ 1 ATA controller, 500,000-hour MTTF ■ 1 power supply, 200,000-hour MTTF ■ 1 fan, 200,000-hour MTTF ■ 1 ATA cable, 1,000,000-hour MTTF Using the simplifying assumptions that the lifetimes are exponentially distributed and that failures are independent, compute the MTTF of the system as a whole.	05	L3	1
	(b)	Discuss Amdahl's law along with the factors that are needed to find the speedup from enhancement.	05	L2	1
4		Define loop unrolling. For the given code snippet Show loop unrolled for four copies of the loop body. Schedule the unrolled loop for the pipeline with the latencies(give in table below). Eliminate any obviously redundant computations and do not reuse any of the registers. Determine the total clock cycles needed to execute the loop for both cases	10	L4	3

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		Loop: L.D F0,0(R1) ; ADD.D F4,F0,F2 ; S.D F4,0(R1) ; DADDUI R1,R1,#-8 ; BNE R1,R2,Loop ;		
5	(a)	Discuss with suitable diagram hardware based speculation for overcoming control dependences.	05	L2
	(b)	With suitable diagram, discuss centralized shared memory architecture.	05	L2

Instruction Producing Result	Instruction using Result	Latency in clock cycles
FP ALU Op	Another FP ALU Op	3
FP ALU Op	Store double	2
Load double	FP ALU Op	1
Load double	Store double	0

COURSE OUTCOMES:

CO1: Understand core concepts of computer architecture to investigate parallel algorithms for computational problems.
CO2: Analyze the performance of parallel programming models and techniques and develop profic parallel algorithm design using MPI, OpenACC, and CUDA.
CO3: Design and implement parallel algorithms using appropriate parallel programming models.
CO4: Demonstrate Parallel computing concepts for suitable compute intensive real time applications

COs/BTL	CO1	CO2	CO3	CO4	L1	L2	L3	L4
Marks	36	--	24	--	1	29	20	10